National University of Computer and Emerging Sciences, Lahore Campus

Course:	Electronics Circuit Design	Course Code:	EE2001
Program:	BS (Electrical Engineering)	Semester:	Fall 2024
Duration:		Total Marks:	20
Submit On:	22 - Oct - 2024	Weight	5
Section:	BEE - 5A	Page(s):	01
Exam:	Assignment - 02	CLO:	02

Question 1:

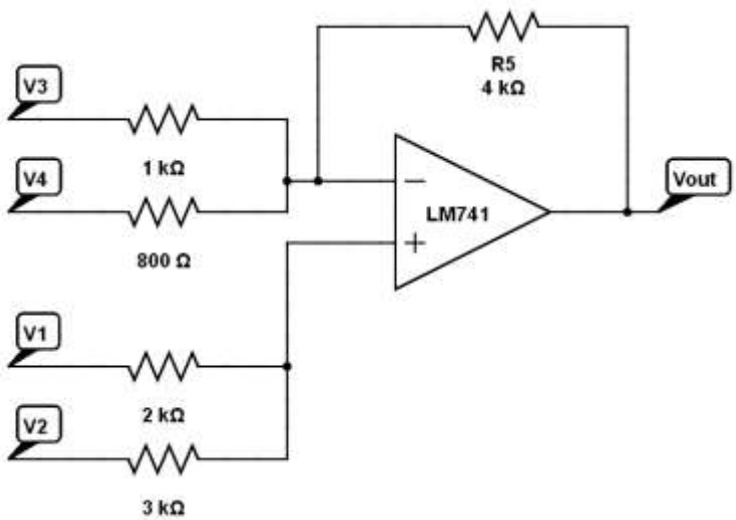
Design a circuit that implements the following function of PID controller using operational-amplifiers. Initiate your design from the two analog input signals, V_{ref} and V_{sense}, and onwards build the rest of the circuit that yields the following output. Use 50 kΩ resistors in the feedback path.

$$V_o = 12(V_e) + 300\left(\int V_e\,dt\right) + 0.5\left(\frac{dV_e}{dt}\right)$$

$$\text{such that,}\qquad V_e = V_{ref} - V_{sensor}$$

Question 2:

a. Analyze the following operational amplifier circuit and find the expression of V_{out}.



b. Implement the function computed in part (a) using the weighted summing amplifier.

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BEE-5A

Q. No. 1

$$V_o = 12(V_e) + 300\left(\int V_e\,dt\right) + 0.5\left(\frac{dV_e}{dt}\right)$$

$\Rightarrow$ $V_e = V_{ref} - V_{sensor}$

$\Rightarrow$ $R_f = 50k\Omega$

Proportional control:-

For non-inverting configuration,

$V_o = \left(1 + \frac{R_f}{R_i}\right)V_i \Rightarrow V_{o1} = G_p V_e$

As G_p (gain of Proportional control) = 12

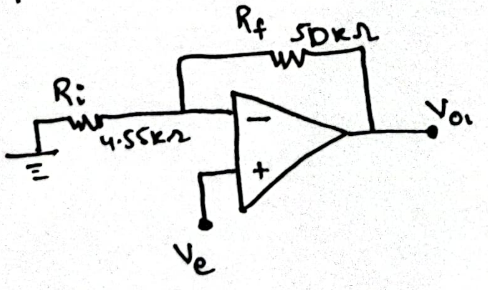
$\Rightarrow$ $12 = 1 + \frac{R_f}{R_i}$

$12 - 1 = \frac{50k}{R_i}$

$R_i = \frac{50k}{11}$

$R_i = 4.55\,k\Omega$

Proportional Control Circuit:-



Integral Control:-

The output voltage of an integrator is given by

$$V_o = -\frac{1}{RC}\int V_i\,dt \Rightarrow V_{o2} = -G_I \int V_e\,dt$$

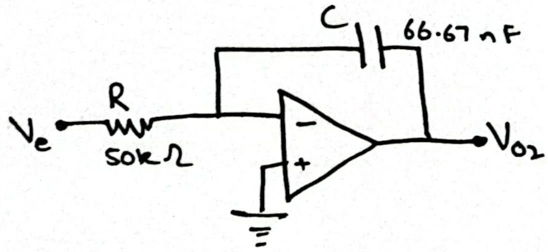
As G_I (gain of Integral Control) = 300

$$\Rightarrow 300 = \frac{1}{RC}$$

$$C = \frac{1}{300(50k)}$$

$$C = 66.67\ nF$$

Integral Control Circuit:-



~~Differen~~ Derivative Control:-

The output voltage of an differentiator is given by

$$V_o = -RC\frac{dV_i}{dt} \Longrightarrow V_{o3} = -G_D\frac{dV_e}{dt}$$

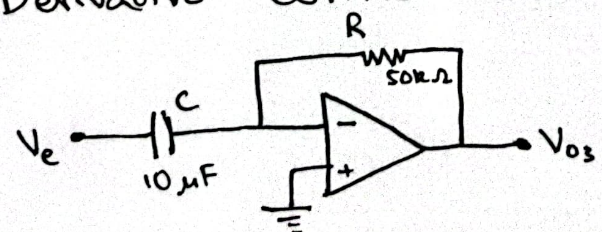
As G_D (gain of ~~the~~ derivative control) = 0.5

$$\Rightarrow 0.5 = RC$$

$$C = \frac{0.5}{50k}$$

$$C = 10\ \mu F$$

Derivative Control Circuit:-



For $V_e = V_{ref} - V_{sensor}$, using a single op-amp difference amplifier

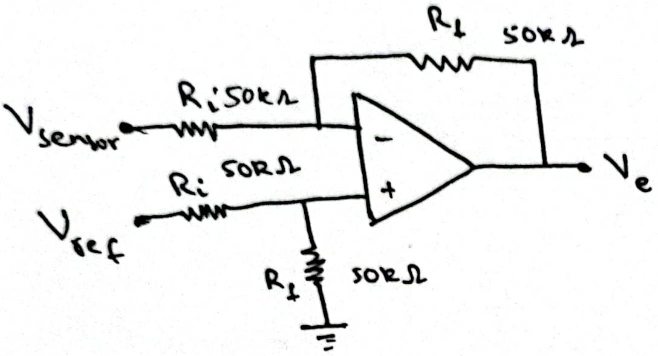
$$V_o = \frac{R_f}{R_i}(V_2 - V_1) \Rightarrow V_e = G(V_{ref} - V_{sensor})$$

As G should be $\Rightarrow 1$

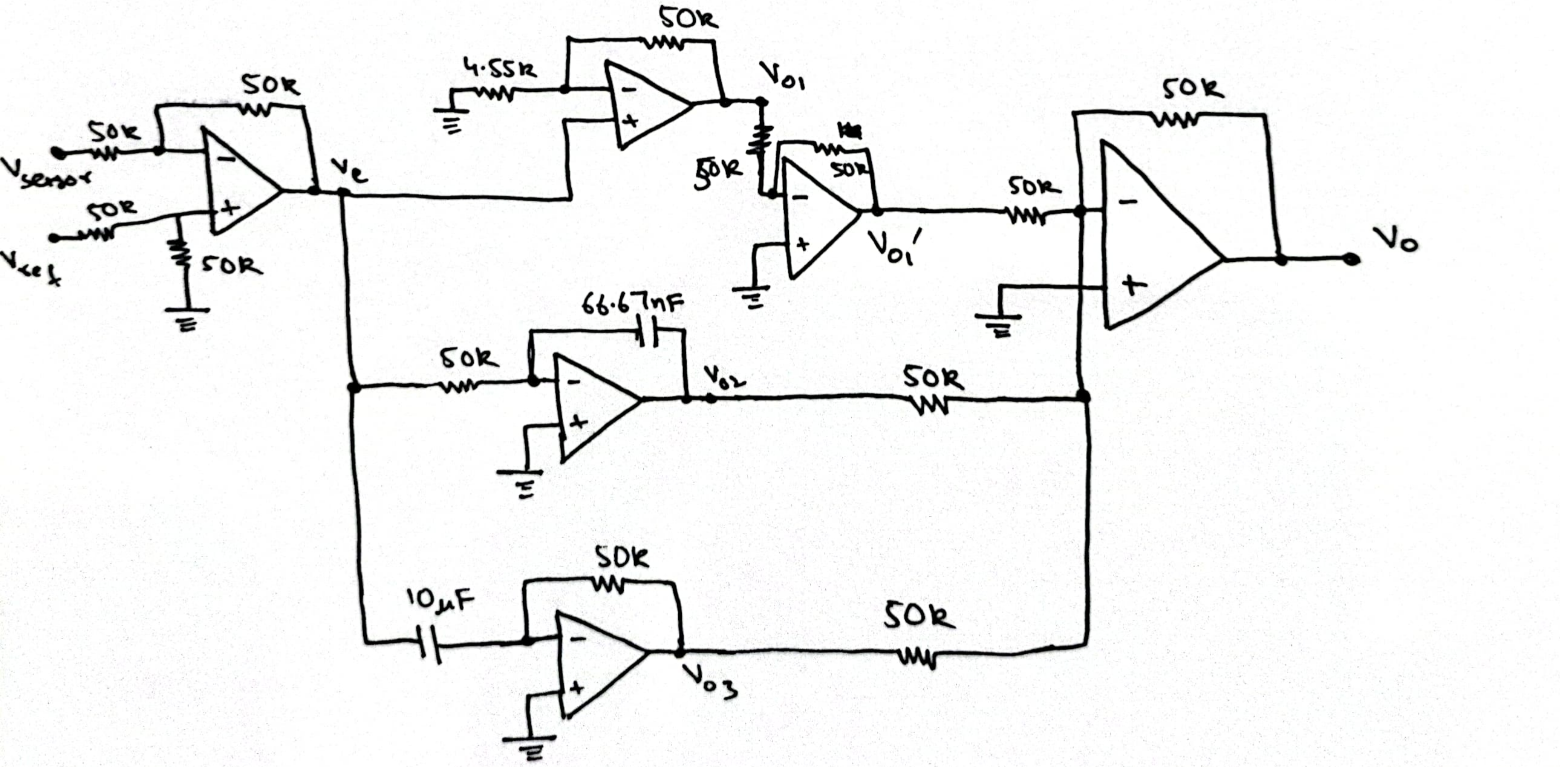
$$\Rightarrow 1 = \frac{R_f}{R_i}$$

$$R_i = \frac{50k}{1}$$

$$R_i = 50k\Omega$$



Combining outputs of individual op-amps using a summing amplifier of gain $G = 1$



Hence, the output voltage is

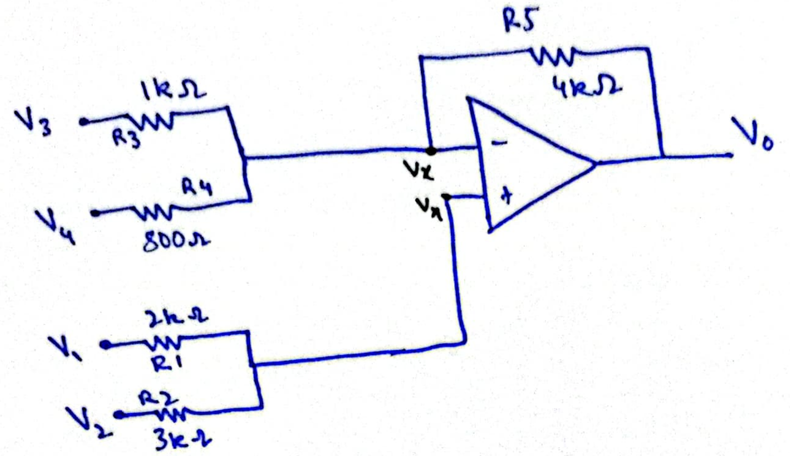
$$V_o = -V_{01}' + V_{02} - V_{03}$$

$$V_o = -(-12V_e) - (-300\int V_e dt) - (-0.5\frac{dV_e}{dt})$$

$$= 12V_e + 300\int V_e\, dt + 0.5\frac{dV_e}{dt}$$

Q. No. 2

(a)



$\Rightarrow \quad V_+ = V_- = V_x$ (internal short circuit)

$$\Rightarrow \quad \frac{V_x - V_1}{2k} + \frac{V_x - V_2}{3k} = 0$$

$$3kV_x - 3kV_1 + 2kV_x - 2kV_2 = 0$$

$$5kV_x - 3kV_1 - 2kV_2 = 0$$

$$5\cancel{k}V_x = (3V_1 + 2V_2)\cancel{k}$$

$$5V_x = 3V_1 + 2V_2$$

$$V_x = \frac{1}{5}(3V_1 + 2V_2) \quad — ①$$

$$\Rightarrow \quad \frac{V_x - V_3}{1k} + \frac{V_x - V_4}{800} + \frac{V_x - V_o}{4k} = 0$$

$$\frac{V_x}{1k} - \frac{V_3}{1k} + \frac{V_x}{800} - \frac{V_4}{800} + \frac{V_x}{4k} - \frac{V_o}{4k} = 0$$

$$\frac{V_x}{400} = \frac{V_3}{1k} + \frac{V_4}{800} + \frac{V_o}{4k}$$

$$V_x = 0.4V_3 + 0.5V_4 + 0.1V_o \quad — ②$$

From ① & ②

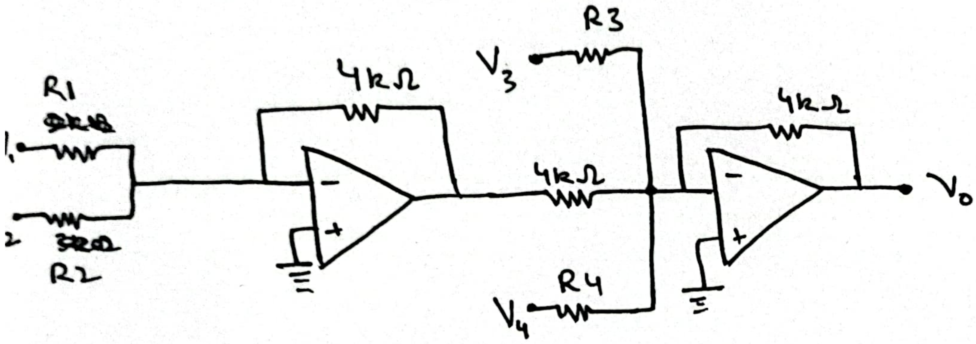
$$\frac{1}{5}(3V_1 + 2V_2) = 0.4V_3 + 0.5V_4 + 0.1V_o$$

$$0.1V_o = 0.6V_1 + 0.4V_2 - 0.4V_3 - 0.5V_4$$

$$\boxed{V_o = 6V_1 + 4V_2 - 4V_3 - 5V_4}$$

(b)

Implementing V_o using a weighted summing amplifier,



For V_1,

$$\frac{R_f}{R_1} = 6$$

$$\frac{4k}{6} = R_1$$

$R_1 = 666.67\Omega$

For V_2,

$$\frac{R_f}{R_2} = 4$$

$$\frac{4k}{4} = R_2$$

$R_2 = 1k\Omega$

For V_3,

$$\frac{R_f}{R_3} = 4$$

$$\frac{4k}{4} = R_3$$

$R_3 = 1k\Omega$

For V_4,

$$\frac{R_f}{R_4} = 5$$

$$\frac{4k}{5} = R_4$$

$R_4 = 800\Omega$